Homeostasis & Insulin

Exam code: 0610 & 0970

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Role of insulin in homeostasis

- Homeostasis is defined as:

  the maintenance of a constant internal environment

- Homeostasis ensures that **internal conditions** within the body (such as temperature, blood pressure, water concentration, glucose concentration etc) are **kept within set limits** in order to ensure that reactions in body cells can function

The role of insulin

- Insulin is secreted into the blood at times when blood glucose levels are **high**
- This is (most often) directly after a meal
- The kidneys can only cope with a certain level of glucose in the blood
  - If the level gets too high, glucose gets **excreted** and is **lost** in the urine
  - This is like running a car with a hole in the petrol tank; valuable fuel is being wasted
- To avoid this, insulin temporarily converts excess glucose into glycogen in the liver and muscles
  - Insulin **decreases blood glucose concentration**

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